UMC 110nm RFCMOS Al Advanced Enhancement Process 1P6M2T40K Psub Inductor SPICE DOCUMENT

Version 0.1 Phase 1
2013/09/12

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Owner	Comment
V0.1p1	09/12/2013	Ji Wei Hsu	Original release of models for 0.8um and 4.0um metal6 inductors based on EM simulation

1.2 General Information

This document describes RF (Radio Frequency) inductor SPICE models. The planar inductors (spiral and symmetric) are built on the metal 5~6 and stack inductor is built on metal 1~4 of UMC's L110 process.

The compact model is provided in various formats. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

The inductor models in this document are generated based on intensive EM (Electromagnetic) simulation. The EM simulation is well-calibrated based DSM data to ensure EM simulation accurately predicts capacitor performance.

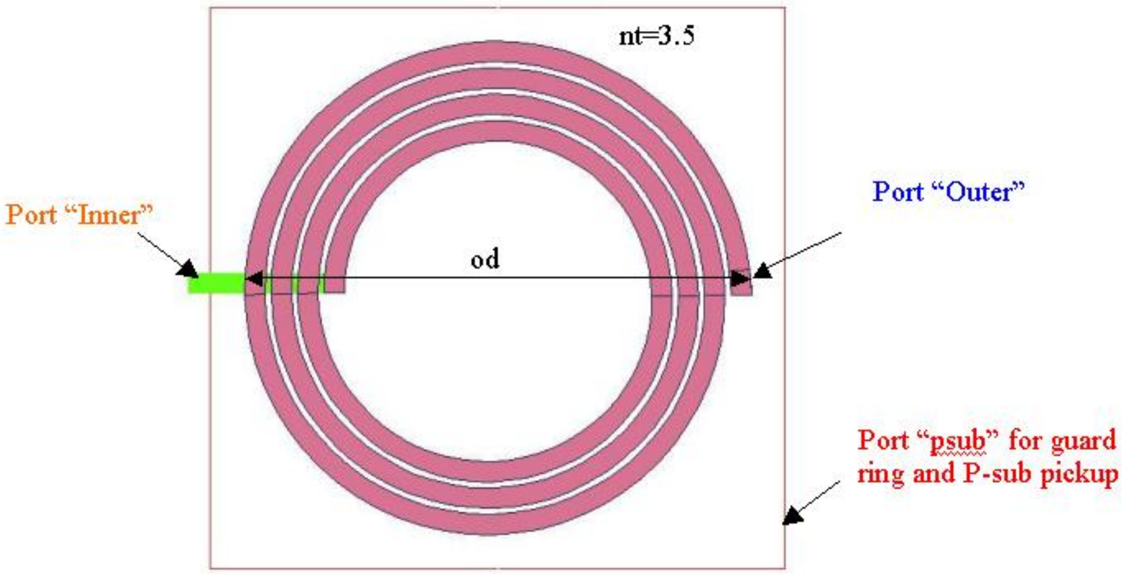
The models should be used in conjunction with UMC capacitor Pcells (Programmable layout Cell) provided in UMC MM/RF FDK (Foundry Design Kit). The models may not be valid if the capacitor layout is different from UMC Pcell.

In current release, 3 major commercial simulators (Cadence's SPECTRE, Synopsys' HSPICE and Mentor Graphics' ELDO) are supported.

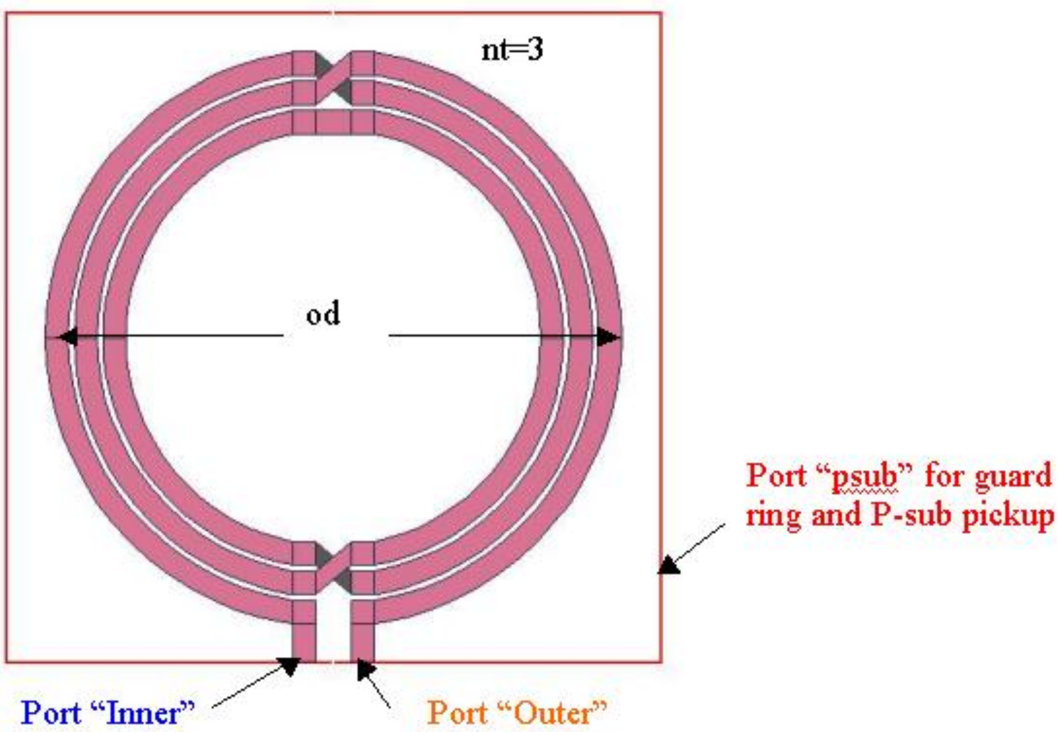
It is suggested to read the model document firstly to understand the model usage and limitation before using the models.

1.3 Device Structure

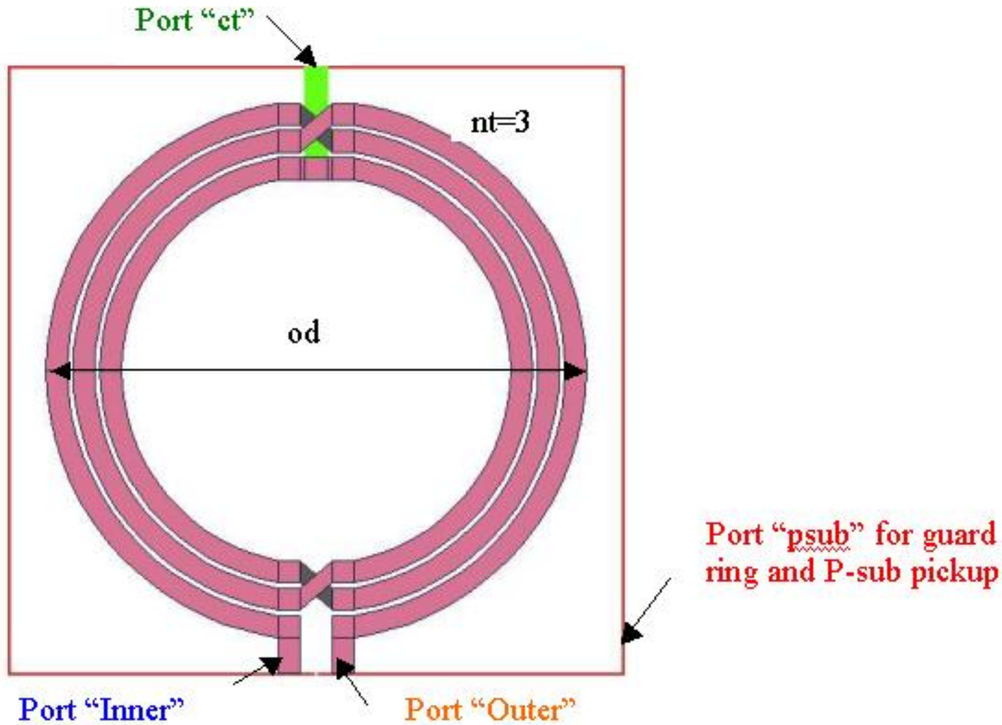
The planar inductors are built on the top 2 metal layers (metal 5~6) to reduce the resistance and the coupling to substrate. The stack inductors are built on metal 1~4 to increase the number of windings.



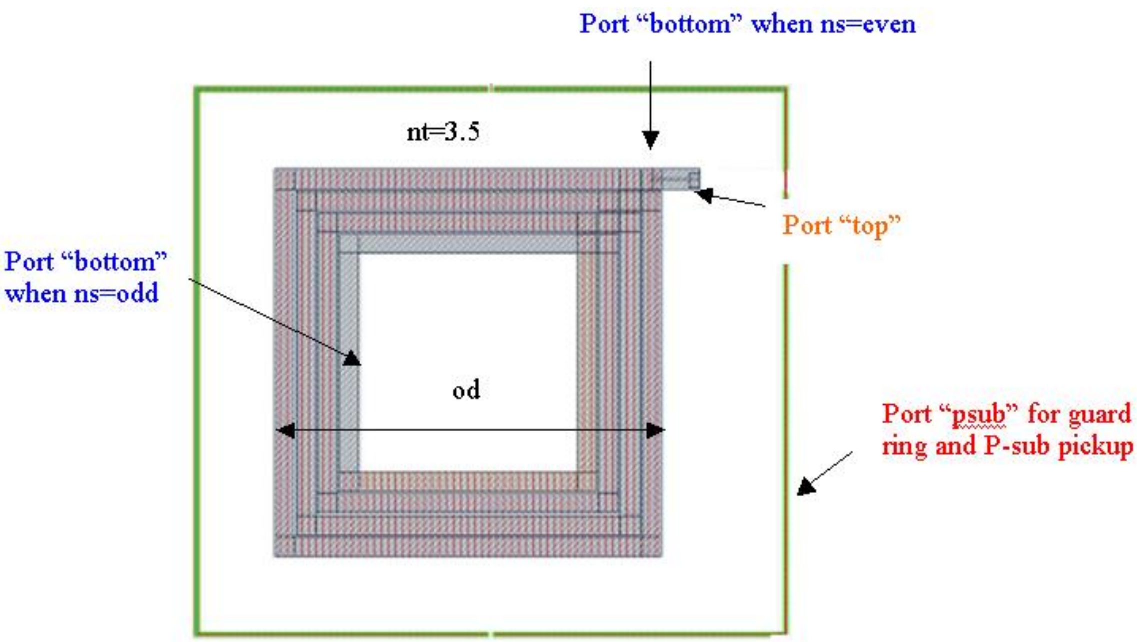
(a) Example of spiral inductor



(b) Example of symmetric inductor



(c) Example of symmetric inductor with center tap



(d) Example of stack inductor

Figure 1-1 Sample layout of Inductor

The device with guard ring is enclosed by CAD layer "RFSYMBOL".

1.4 Model Usage

It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of simulator might be different.

Synopsys HSPICE release 2012.06
Cadence SPECTRE version 10.1.1.111.isr8 32bit
Mentor Graphics ELDO version 6.11_1.1

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below.

FF: minimum Q value
TT: typical Q value
SS: maximum Q value

Follow the description for net-list of inductors, and these steps to invoke the model file during simulation.

For Hspice
-Copy ModelFileName.lib & ModelFileName.mdl into your simulation directory.
-Add the following statement in the input netlist to the simulator
.lib "./ModelFileName.lib" CornerSection
, where CornerSection could be FF, TT, SS.
-Follow the example below to call the model

Example 1: For Spiral Inductor (Refer to Figure 1-1 (a))
x1 outer inner psub DeviceName od=200u w=8u s=2.3u nt=4.5

Example 2: For Symmetric Inductor (Refer to Figure 1-1 (b))
x1 outer inner psub DeviceName od=200u w=8u s=2.3u nt=4

Example 3: For Symmetric Inductor with Center Tap (Refer to Figure 1-1 (c))
x1 outer inner psub DeviceName od=200u w=8u s=2.3u nt=4

Example 4: For Stack Inductor (Refer to Figure 1-1 (d))
x1 top bottom psub DeviceName od=45u w=2u s=0.28u nt=4.5 bm=3 ns=2.5

For Cadence Spectre
-Copy ModelFileName.lib.scs & ModelFileName.mdl.scs into your simulation directory.
-Add the following statement in the input netlist to the simulator
include "./ModelFileName.lib.scs" section=CornerSection
, where CornerSection could be FF, TT, SS.

-Follow the example below to call the model

Example 1: For Spiral Inductor (Refer to Figure 1-1 (a))
x1 (outer inner psub) DeviceName od=200u w=8u s=2.3u nt=4.5

Example 2: For Symmetric Inductor (Refer to Figure 1-1 (b))
x1 (outer inner psub) DeviceName od=200u w=8u s=2.3u nt=4

Example 3: For Symmetric Inductor with Center Tap (Refer to Figure 1-1 (c))
x1 (outer inner psub) DeviceName od=200u w=8u s=2.3u nt=4

Example 4: For Stack Inductor (Refer to Figure 1-1 (d))
x1 (top bottom psub) DeviceName od=45u w=2u s=0.28u nt=4.5 bm=3 ns=2.5

For Mentor Graphics Eldo
-Copy ModelFileName.lib.eldo & ModelFileName.mdl.eldo into your simulation directory.
-Add the following statement in the input netlist to the simulator
.lib "./ModelFileName.lib.eldo" CornerSection
, where CornerSection could be FF, TT, SS.
-Follow the example below to call the model

Example 1: For Spiral Inductor (Refer to Figure 1-1 (a))
x1 outer inner psub DeviceName od=200u w=8u s=2.3u nt=4.5

Example 2: For Symmetric Inductor (Refer to Figure 1-1 (b))
x1 outer inner psub DeviceName od=200u w=8u s=2.3u nt=4

Example 3: For Symmetric Inductor with Center Tap (Refer to Figure 1-1 (c))
x1 outer inner psub DeviceName od=200u w=8u s=2.3u nt=4

Example 4: For Stack Inductor (Refer to Figure 1-1 (d))
x1 top bottom psub DeviceName od=45u w=2u s=0.28u nt=4.5 bm=3 ns=2.5

DeviceName & ModelFilename Table

As stated above that one scalable model is provided and this model is valid within the following geometry ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

Geometry range:
For Spiral Inductors:
Outer diameter (od): 75 ~ 299 um (Before shrinkage)
metal width (w): 2.3 ~ 9.7 um (Before shrinkage)
metal spacing (s): 2.3 ~ 5 um (Before shrinkage)
number of turn (nt): 1.5, 2.5, 3.5, 4.5, 5.5, 6.5, 7.5

For Symmetric Inductors:

Outer diameter (od): 75 ~ 299 um (Before shrinkage)
metal width (w): 2.3 ~ 9.7 um (Before shrinkage)
metal spacing (s): 3.2 ~ 5 um (Before shrinkage)
number of turn (nt): 2 ~ 7 (Integer only)

For Symmetric with center tap Inductors:
Outer diameter (od): 75 ~ 299 um (Before shrinkage)
metal width (w): 2.23 ~ 9.7 um (Before shrinkage)
metal spacing (s): 2.23 ~ 5 um (Before shrinkage)
number of turn (nt): 2 ~ 7 (Integer only)

For Stack Inductors:
Outer diameter (od): 45 ~ 75 um (Before shrinkage)
metal width (w): 2 ~ 3 um (Before shrinkage)
metal spacing (s): 0.28 ~ 0.4 um (Before shrinkage)
number of turn (nt): 1.5, 2.5, 3.5, 4.5, 5.5
bottom metal layer (bm): 1, 2, 3
number of metal stack (ns): 2 ~ (5- bm) . if bm=1, ns can be “even” only for routing issue

Geometry constrains:
Because this model should be used in conjunction with UMC transformer Pcells provided in UMC MM/RF FDK. In order to make transformer layout no DRC error, the following rule needs to be satisfied.
For Spiral Inductors:

$$od - 2*(nt-0.5)*s - 2(nt+0.5)*w > 30e-6$$

For Symmetric Inductors:

$$\begin{cases} \dfrac{\frac{od}{2} - nt*w - (nt-1)*s}{\sqrt{2}} \geq \dfrac{3*w+2*s}{2} \\ \sqrt{[\frac{od}{2} - (nt-1)*(w+s)]^2 - [\frac{3*w+2*s}{2}]^2} - \sqrt{[\frac{od}{2} - (nt-1)*(w+s) - w]^2 - [\frac{3*w+2*s}{2}]^2} < 12e-6 \end{cases}$$

For Stack Inductors:

$$od - 2*(nt-0.5)*s - 2(nt+0.5)*w > 10e-6$$

Frequency range:
100 MHz ~ 0.8*Resonance (Max 20 GHz)

2 Inductors

2.1 Electrical Parameters

The important electrical parameters for the characterization of this device are summarized below.

For Spiral, Symmetric, and Stack Inductors

$$L(f) = imag(\frac{1}{y11})/(2\cdot\pi\cdot f)$$

$$Q(f) = imag(\frac{1}{y11})/real(\frac{1}{y11})$$

For Symmetric Inductors with Center Tap

$$DifferentialL(f) = imag(Z)/(2\cdot\pi\cdot f)$$

$$DifferentialQ(f) = imag(Z)/real(Z)$$

Where Z = z11+z22-z12-z21

and z11 = y22/det z12 = -y12/det z21 = -y21/det z22 = y11/det

(det = y11*y22-y12*y21)

2.2 Inductor Model

Model verification plots

The simulation results (solid line) obtained from the extracted model were plotted together with the EM results (dotted line) for comparison.

L vs. Frequency and Q vs. Frequency plots

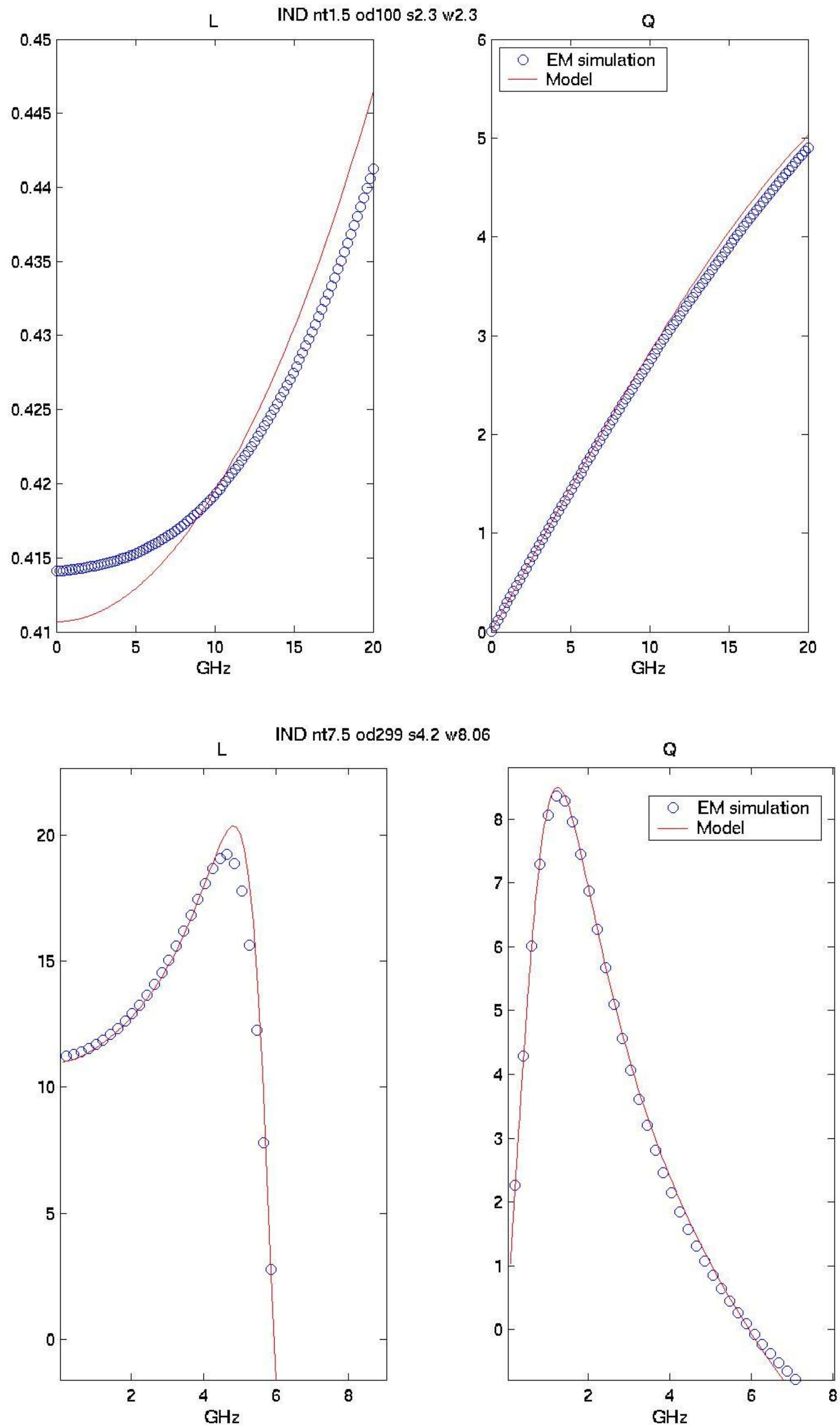


Figure 2-1 LQ plot of Spiral Inductor

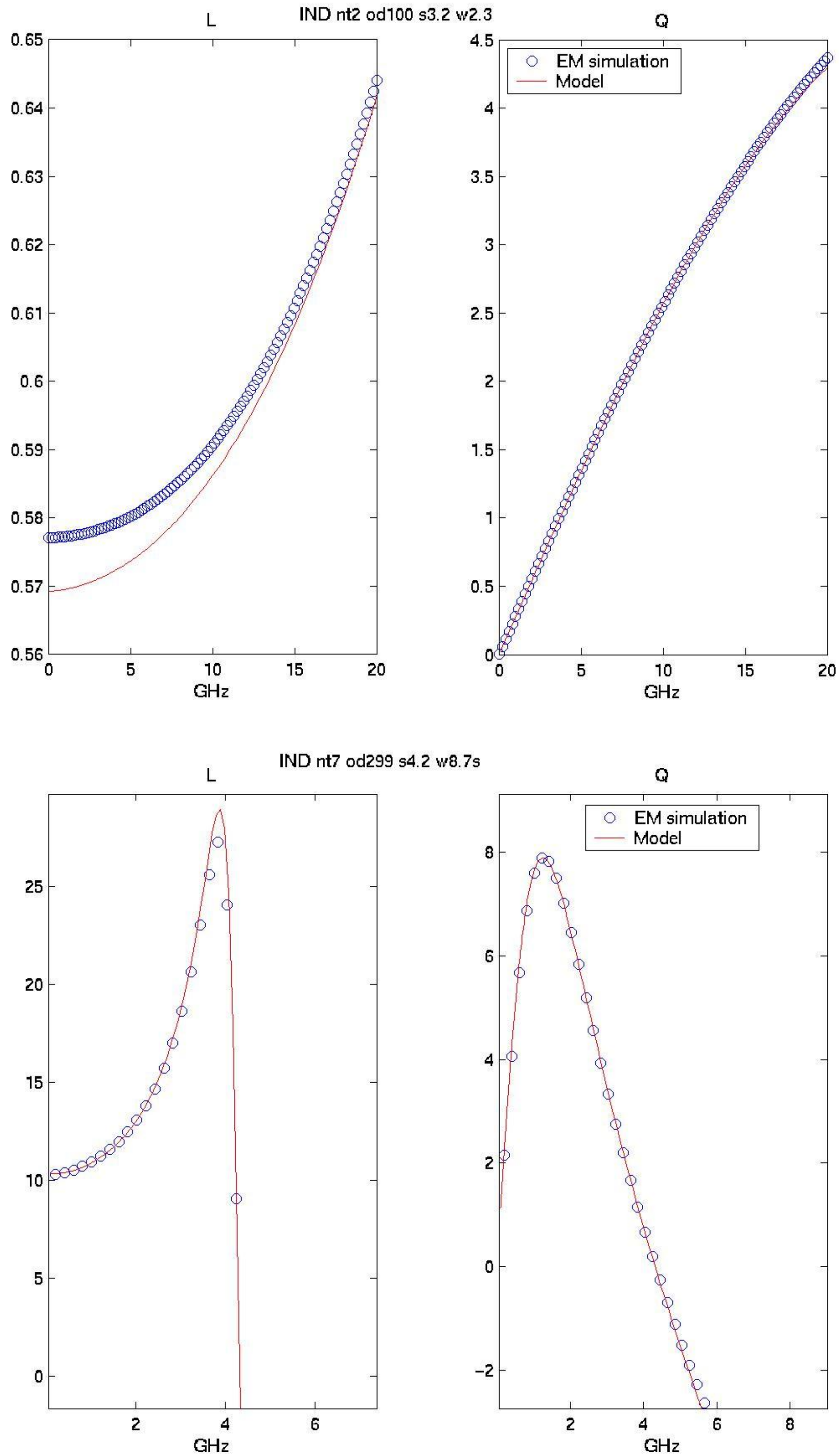


Figure 2-2 LQ plot of Symmetric Inductor

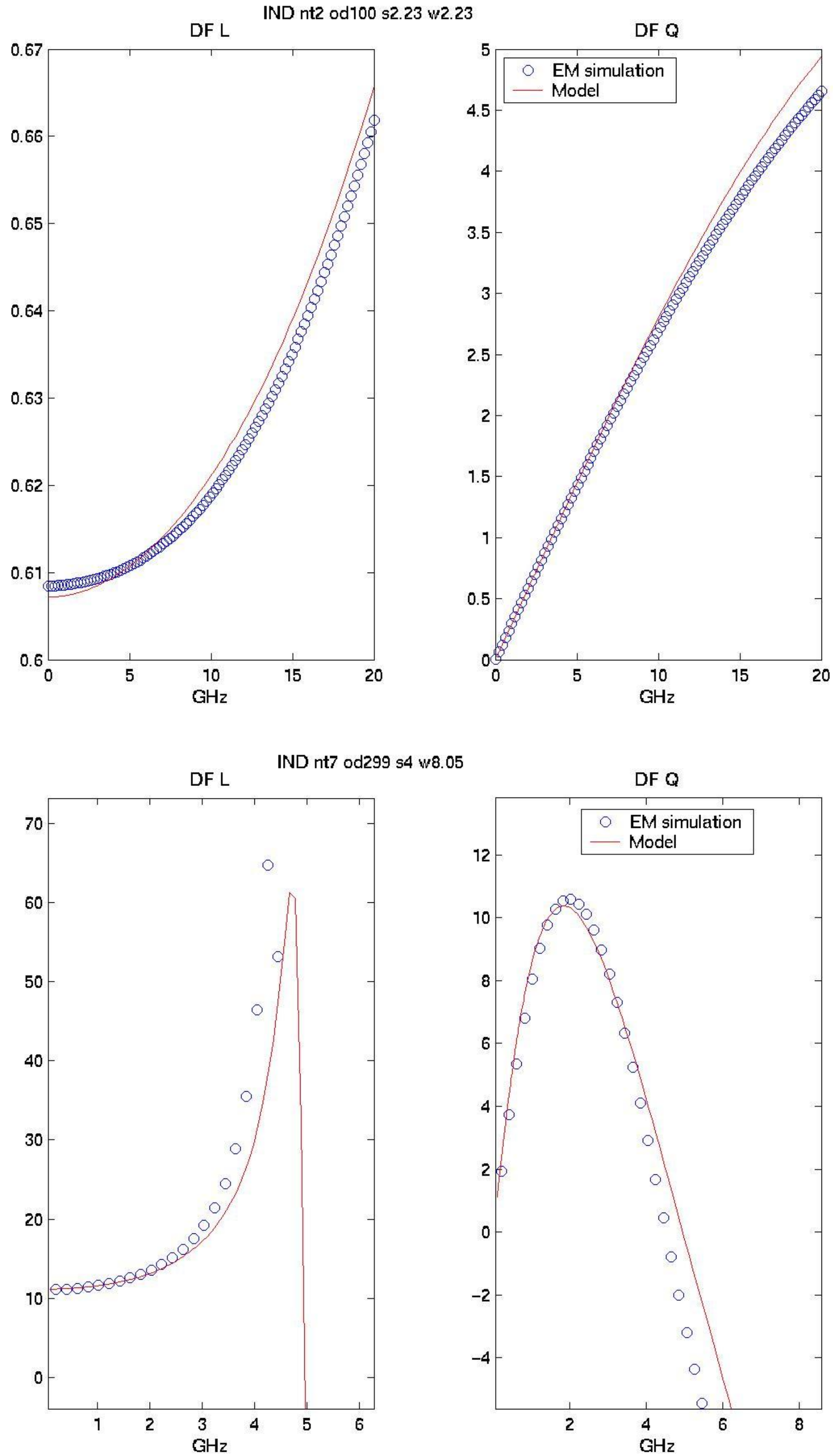
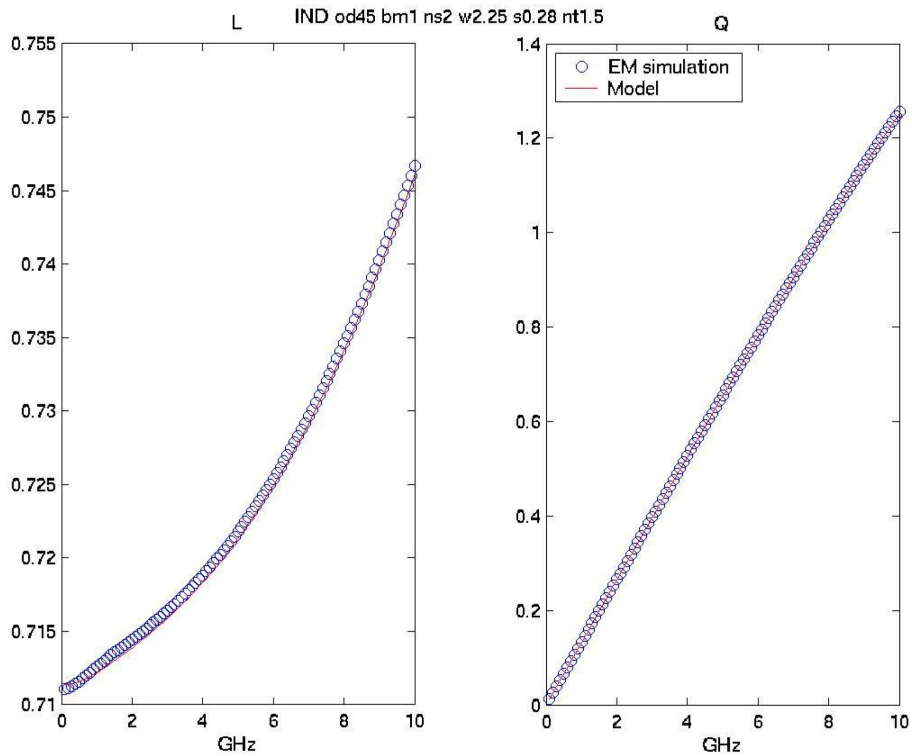


Figure 2-3 LQ plot of Symmetric Inductor with Center Tap



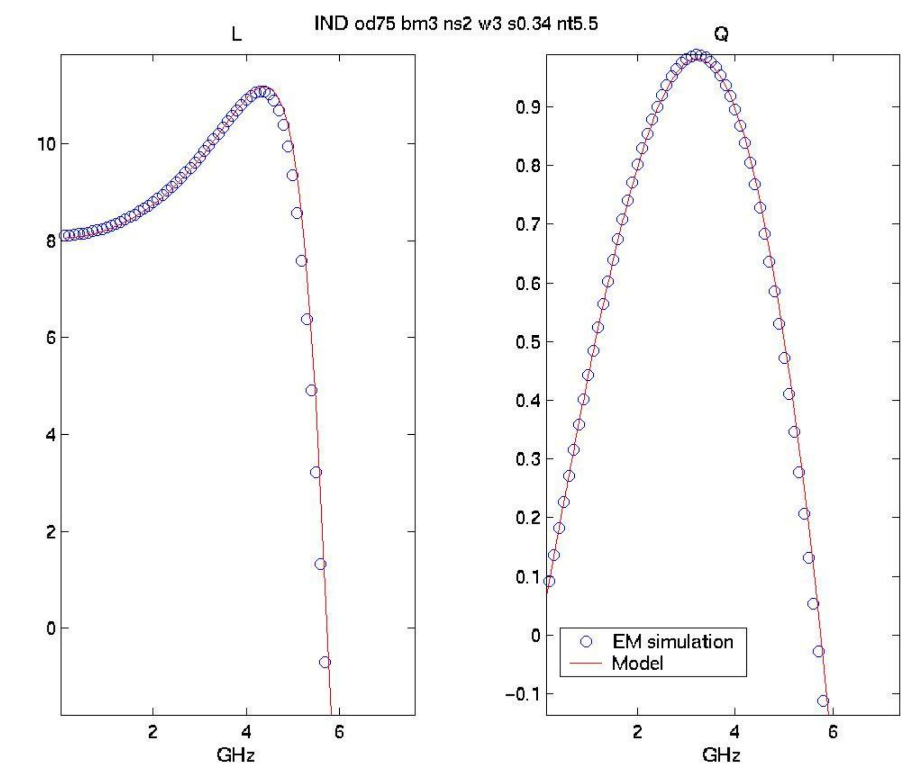


Figure 2-4 LQ plot of Stack Inductor

3 Appendix

3.1 HSPICE Warning Message

Table 3-1 HSPICE warning message

NO.	Warning Message	Explanation
1.	NA	NA

3.2 ELDO Warning Message

Table 3-2 Eldo warning message

NO.	Warning Message	Explanation
1.	NA	NA

3.3 SPECTRE Warning Message

Table 3-3 Spectre warning message

NO.	Warning Message	Explanation
1.	NA	NA